

Hackathon

Bit by Bit Coding - Term 2 | Weeks 9-11

Key Concepts

- Plan before coding: define the goal, features, and who does what.
- A good README explains what the project is and how to run it.
- Git tracks changes; GitHub shares them online.
- Modular code: split the program into small files/functions.
- Present clearly: the problem, a demo, and what you learned.

Syntax Reference

Git Basics

```
git init          # start a new repo
git add .         # stage all changes
git commit -m "add login" # save a snapshot
git push          # upload to GitHub
git status        # see what changed
```

README Structure

```
# Project Title
## What it does
A one-line description.
## How to run
python main.py
## Features
- Login
- Score tracking
## Team
- Ali - backend
- Sara - UI
```

Modular Code Design

- Keep main.py small; put helpers in separate files and import them.
- One feature per commit with a clear message.
- Assign roles so every team member contributes.

Tips & Common Mistakes

- Commit often with clear messages - each commit is a save point.
- Run git status when you're unsure what changed.
- Never push secrets or passwords to GitHub.
- Test the project on a fresh clone before presenting.
- Rubric: functionality, code quality, teamwork, presentation - check each.